

Boson star-black hole binaries: Initial data and head-on collisions

Zhuan Ning (宁专), UCAS

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Supervisor: Rong-Gen Cai

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Outline

- Introduction
- Formalism
- Initial data
- Numerical results
- Conclusions and discussion

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□ Introduction

□ Formalism

□ Initial data

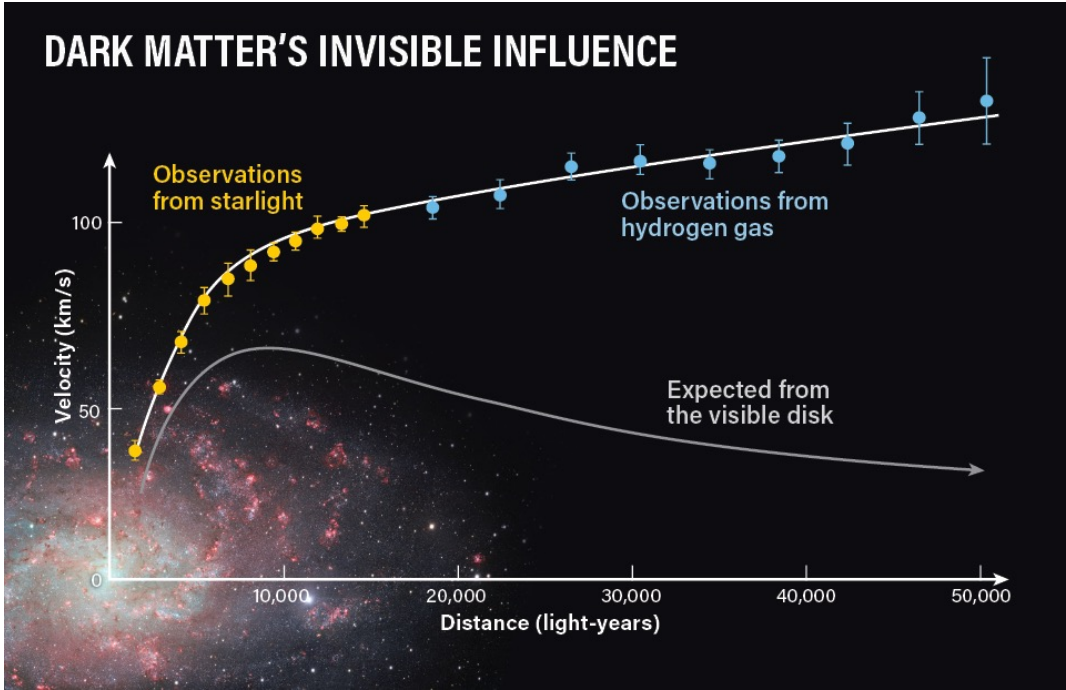
□ Numerical results

□ Conclusions and discussion

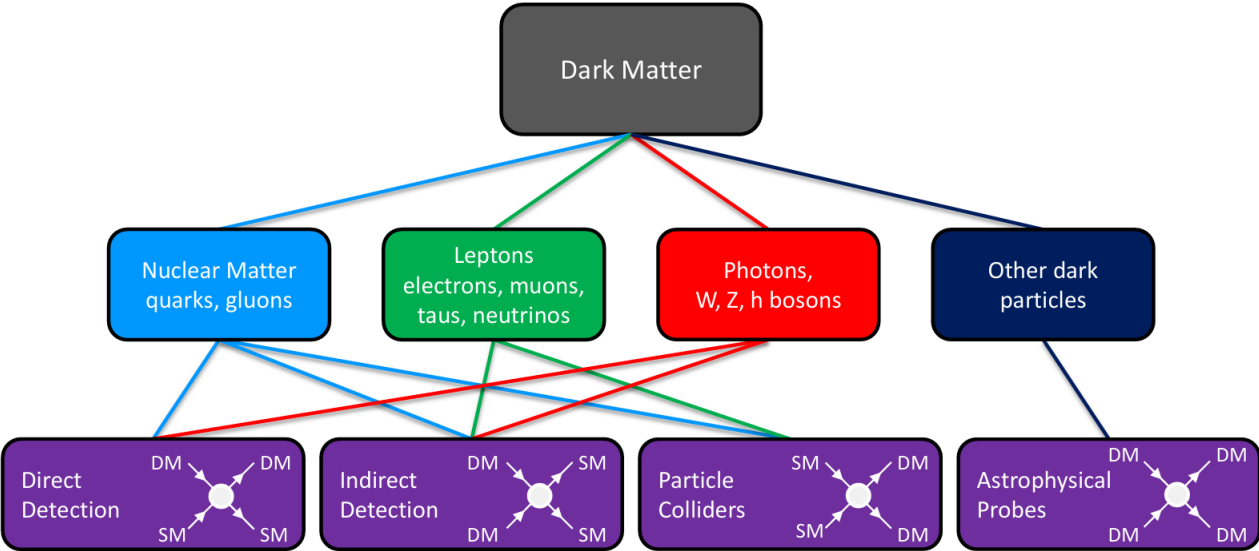
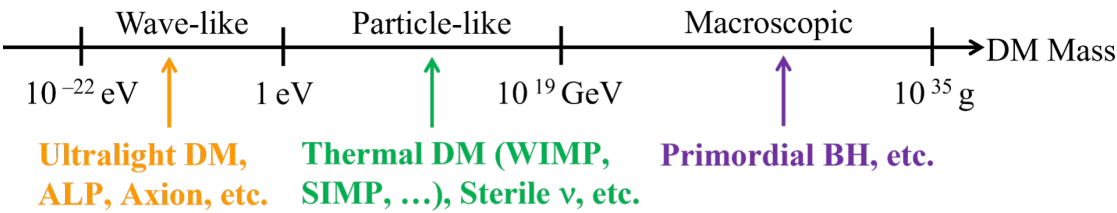
Dark matter

Dark matter (DM): many gravitational observations

They may have non-gravitational interactions, but no conclusive detection!



Many DM candidates:



Scalar-field dark matter

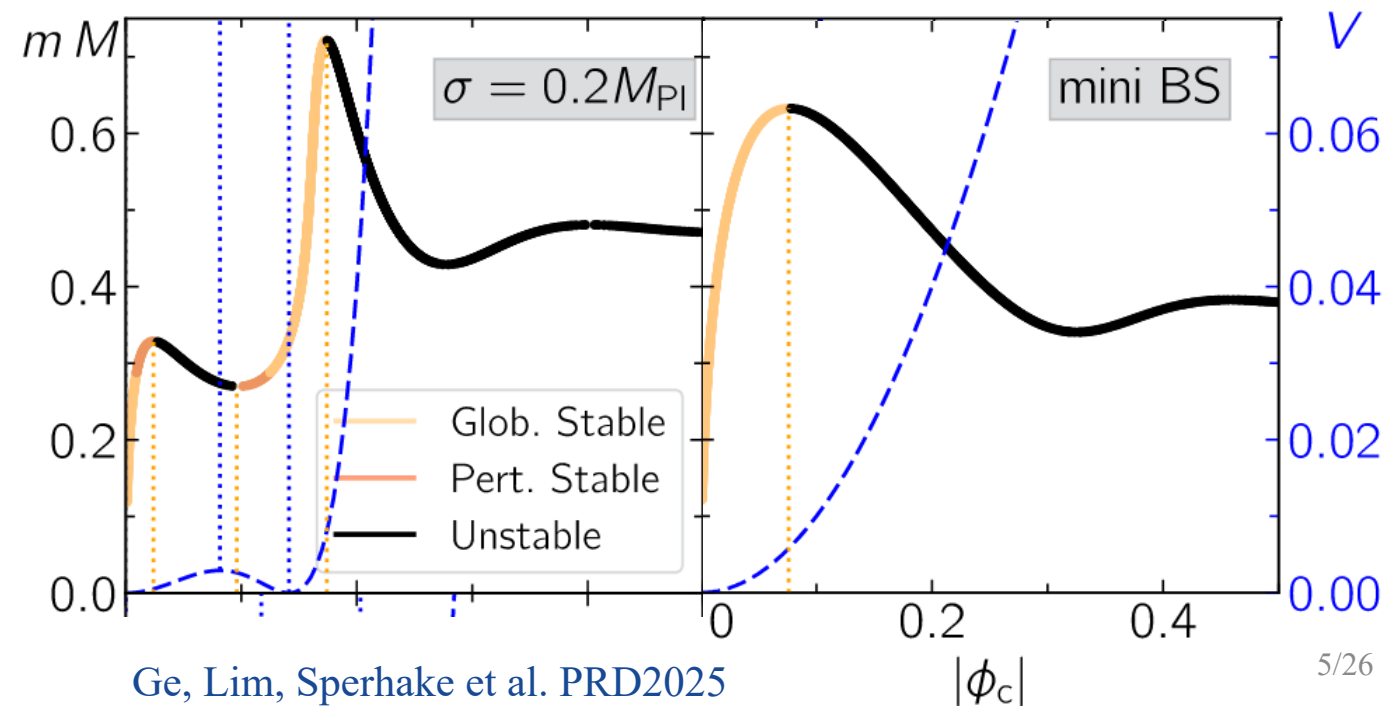
- Scalar-field DM is well-motivated
 - QCD axion
 - axion-like particles from string-theory compactification
 - “axiverse”

- They can form **self-gravitating structures**

- Real scalar: oscillatons
- Complex scalar: boson stars (BSs)

$$M_{\text{max}} \sim M_{\text{pl}}^2 / \mu$$

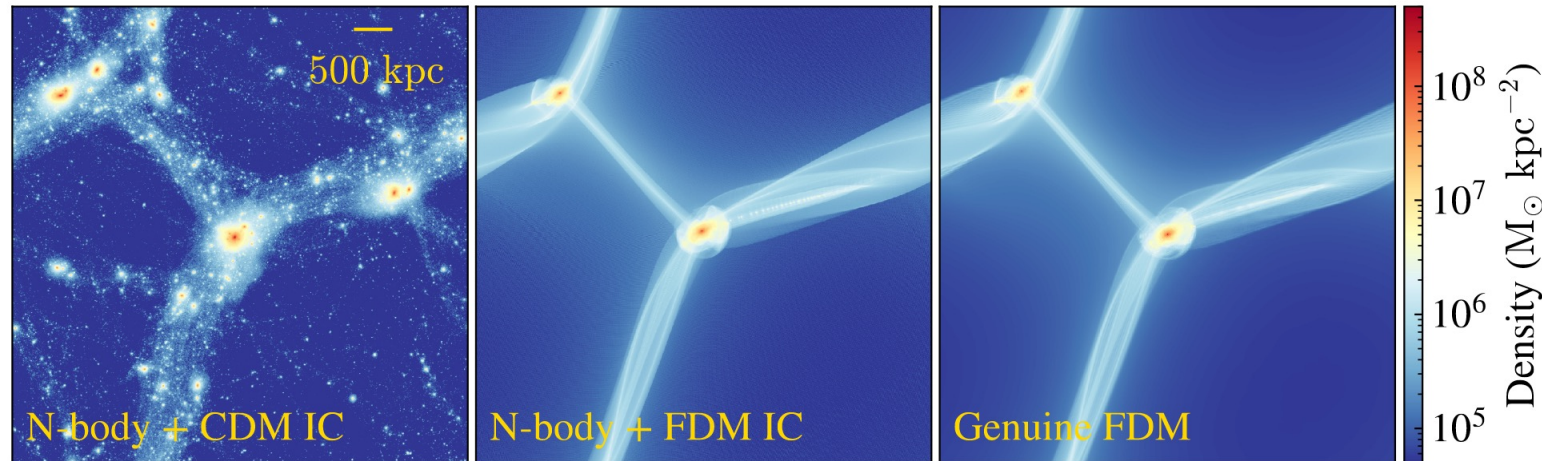
μ is the scalar-field mass



Scalar-field dark matter

Schive 2026
Liv.Rev.Comput.Astrophys

- Fuzzy DM ($\mu \sim 10^{-22} - 10^{-20} \text{eV}$): solitonic cores, kpc-scale



- Heavier scalars ($\mu \sim 10^{-10} - 10^{-16} \text{eV}$): from solar to supermassive masses
- Exotic compact objects (ECOs) or black-hole (BH) mimickers

Their mergers may produce gravitational waves (GWs)

- A new observational window!

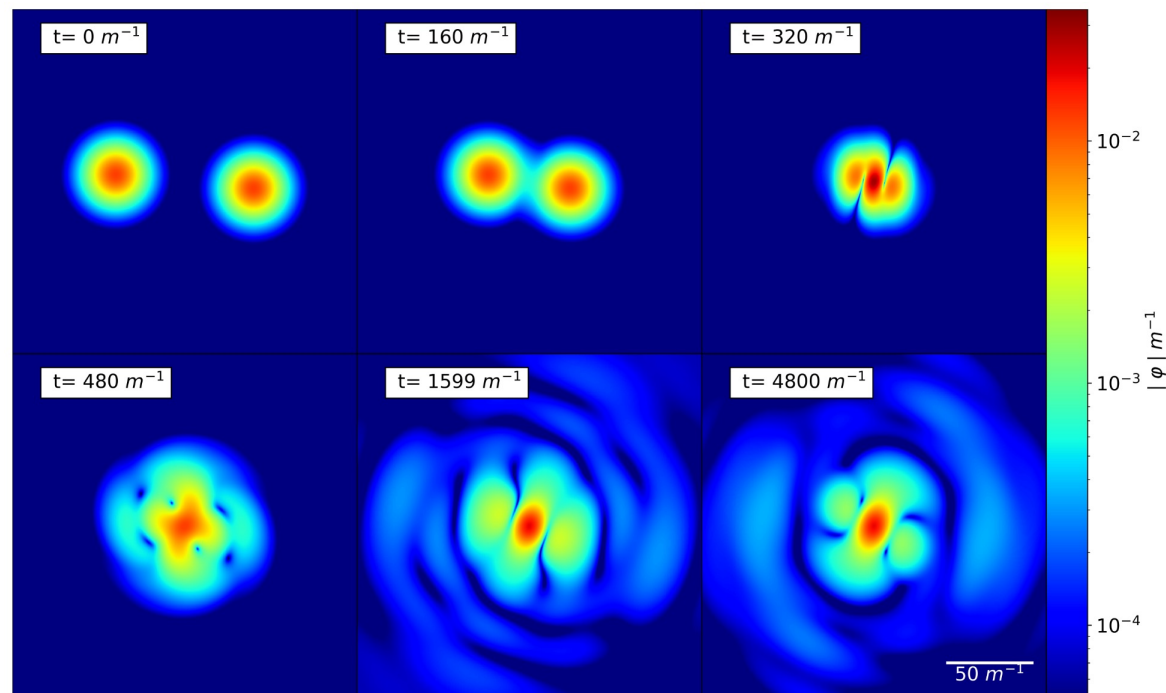
BS-BS binary

➤ Use numerical relativity to study ECO binary mergers and GW signals

- boson star head-on collisions [23, 42, 72, 86, 149, 167, 200, 211]; Cardoso et al. PRD2016 ...
- boson star orbital mergers [47, 49, 50, 95, 129, 210, 212, 240, 251, 252]; Palenzuela et al. PRD2017 ...
- oscillatons (real scalar fields) [148, 265]; Helfer et al. PRD2019, Widdicombe et al. JCAP2020
- ℓ -boson stars [167];
- axionic boson stars [239];
- Proca stars [66, 67, 231, 236]. Sanchis-Gual et al. PRD2019 ...

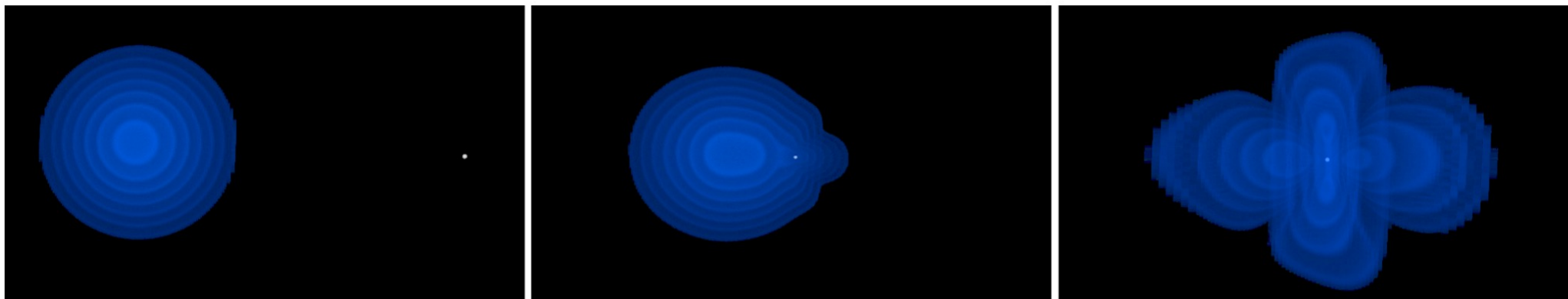
Bezares,
Sanchis-
Gual
2406.04901

Croft, Helfer et al. CQG2023

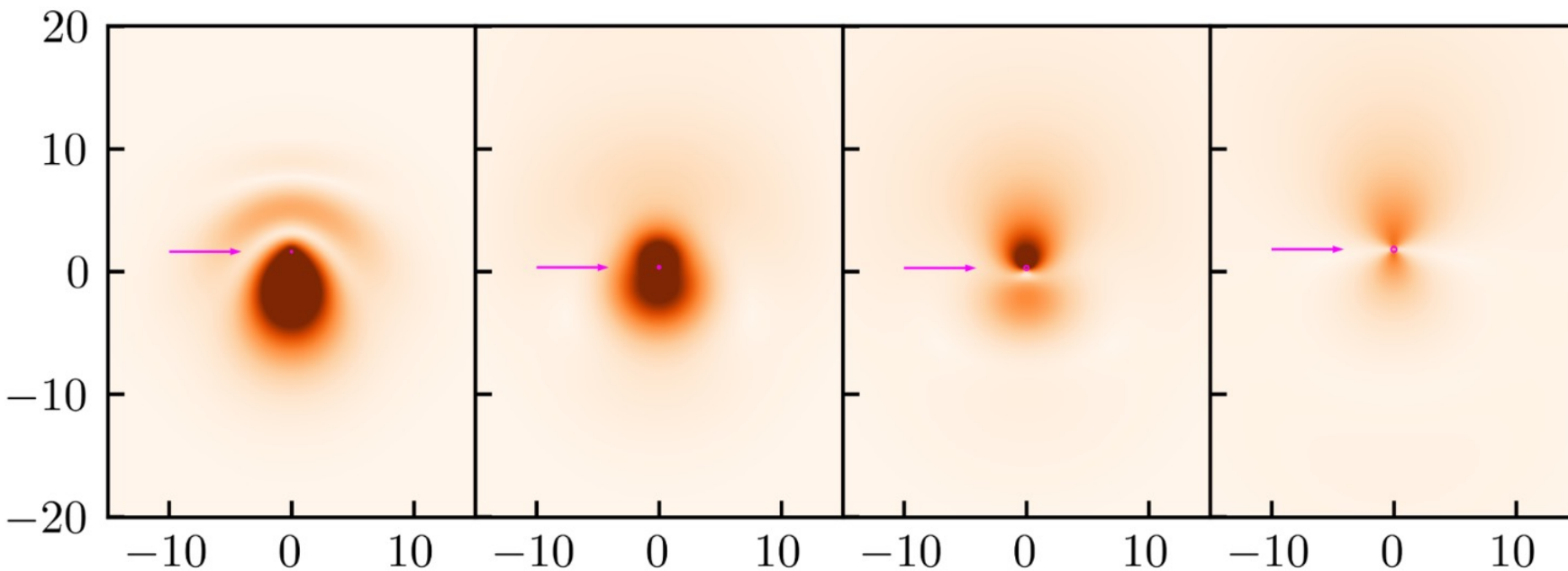


BS-BH binary

- However, encounter of BS and BH is inevitable



Clough, Dietrich, Niemeyer PRD2018: oscillaton-BH head-on collisions

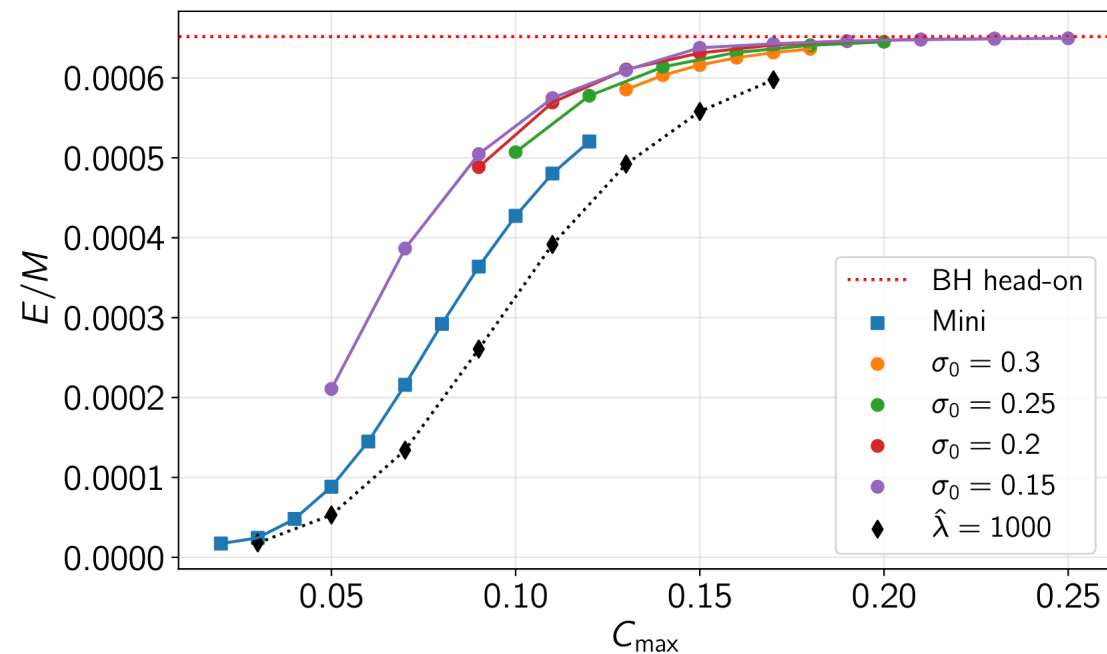
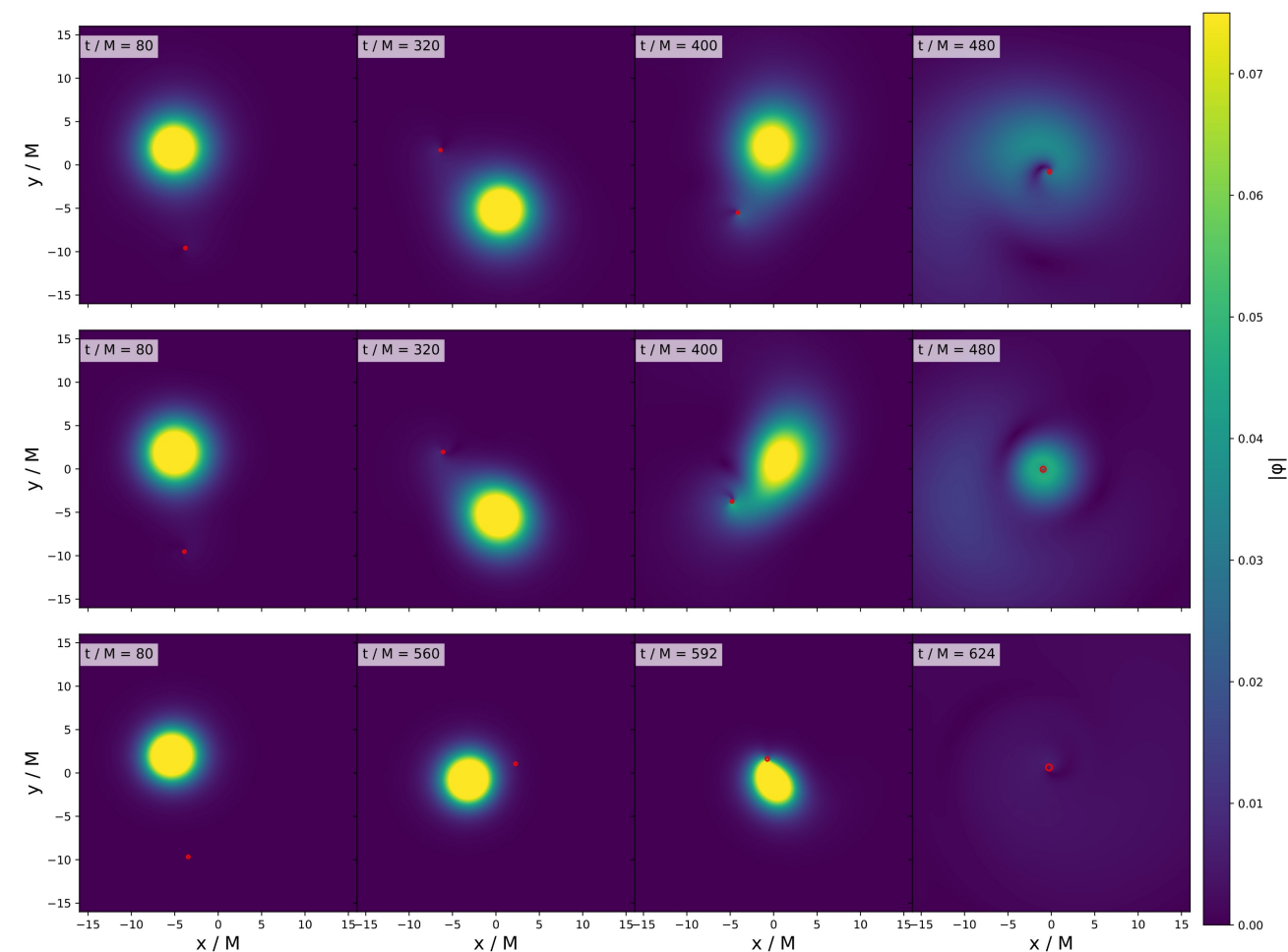


Cardoso, Ikeda, Zhong et al. PRD2022;
Zhong, Cardoso et al. PRD2023
BS-BH head-on collisions
Large mass ratio, $\frac{M_{BS}}{M_{BH}} \sim 10$
BH accretes more than 95% of the BS
material

BS-BH binary

➤ Comparable-mass BS-BH binary: Very recently!

- May produce distinct GW signatures and initial-data quality is critical



BS-BH head-on collisions and inspirals
Marks, Staelens, Sperhake 2604.06312

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Formalism

Model

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi} - \frac{1}{2} (g^{\mu\nu} \nabla_\mu \bar{\varphi} \nabla_\nu \varphi + V(|\varphi|^2)) \right]$$

Solitonic potential

$$V = \mu^2 |\varphi|^2 \left(1 - 2 \frac{|\varphi|^2}{\sigma_0^2} \right)^2$$

$$\sigma_0 = 0.2$$

Noether charge

$$J^\mu = \frac{i}{2} (\varphi \nabla^\mu \bar{\varphi} - \bar{\varphi} \nabla^\mu \varphi),$$

$$N = \int d^3x \sqrt{-g} J^0,$$

3 + 1 formulation

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt)$$

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij},$$

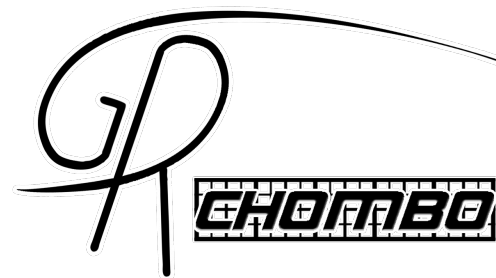
$$\begin{aligned} \partial_t K_{ij} = & -D_i D_j \alpha + \alpha (\mathcal{R}_{ij} + K K_{ij} - 2K_{ik} K^k_j \\ & + 4\pi [(S - \rho) \gamma_{ij} - 2S_{ij}]) + \mathcal{L}_\beta K_{ij}, \end{aligned}$$

$$\partial_t \varphi = -\alpha \Pi + \mathcal{L}_\beta \varphi,$$

$$\begin{aligned} \partial_t \Pi = & \alpha (K \Pi + V' \varphi - D_i D^i \varphi) \\ & - (D_i \alpha) D^i \varphi + \mathcal{L}_\beta \Pi. \end{aligned}$$

$$\mathcal{H} \equiv \mathcal{R} + K^2 - K_{ij} K^{ij} - 16\pi \rho = 0,$$

$$\mathcal{M}_i \equiv D_j K^j_i - D_i K - 8\pi j_i = 0,$$



BSSN/CCZ4
+AMR

$$\Psi_4^{\ell m}(t, R_{\text{ex}}) = \int_{S^2} \Psi_4(t, R_{\text{ex}}, \theta, \phi) \overline{Y_{\ell m}^{-2}}(\theta, \phi) \sin \theta d\theta d\phi,$$

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Isolated boson star and black hole

- Isotropic coordinates

$$ds^2 = -\alpha^2(r)dt^2 + \psi^4(r)\delta_{ij}dx^i dx^j$$

- Boson star

$$\varphi(t, r) = A(r)e^{i\omega t}$$

Specifying $\varphi_{ctr} = A(0) \longrightarrow \alpha(r)$, $\psi(r)$, and $A(r)$

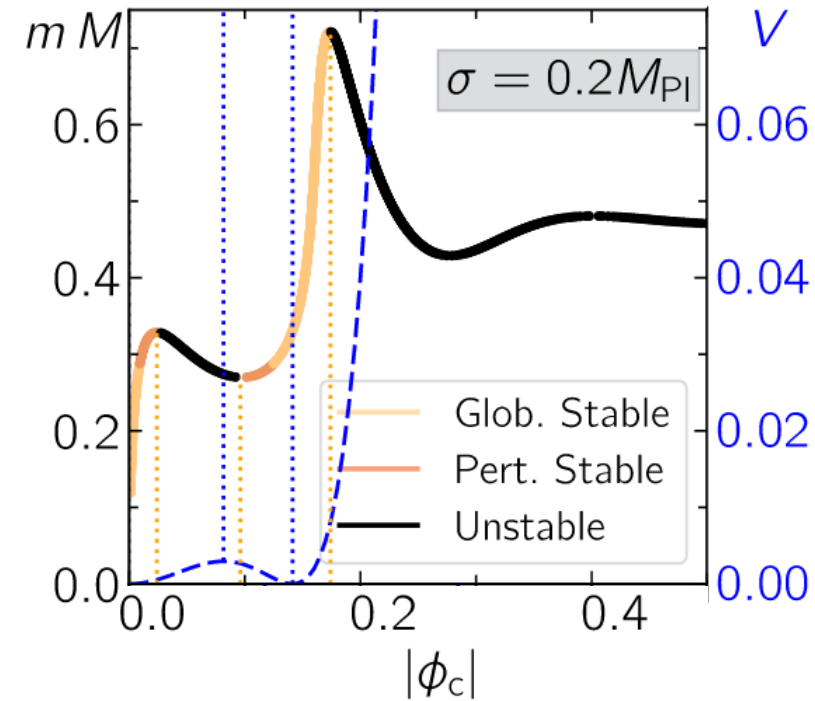
with asymptotically flat boundary conditions

- Black hole

$$\alpha_{BH}(r) = \frac{1 - M_{BH}/2r}{1 + M_{BH}/2r},$$

$$\psi_{BH}(r) = 1 + \frac{M_{BH}}{2r},$$

with vanishing scalar field



Ge, Lim, Sperhake et al. PRD2025

BS, BH $\longrightarrow$ Lorentz boost to velocity v

Plain superposition and its malaise

Plain superposition

$$\gamma_{ij} = \gamma_{ij}^A + \gamma_{ij}^B - \delta_{ij},$$

$$K_{ij} = \gamma_{m(i} \left[K_{j)n}^A \gamma_A^{mn} + K_{j)n}^B \gamma_B^{nm} \right],$$

$$\varphi = \varphi_A + \varphi_B,$$

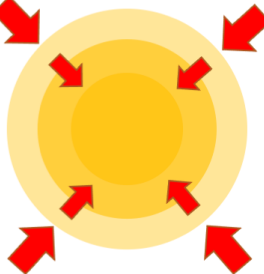
$$\Pi = \Pi_A + \Pi_B.$$

Fos BS-BS binary, perturbation $\delta\gamma_{ij} = \gamma_{ij}^B(x_A^i) - \delta_{ij}$



$$\gamma_{ij} = \gamma_{ij}^A + \gamma_{ij}^B - \delta_{ij}$$

- Squeeze through initial data



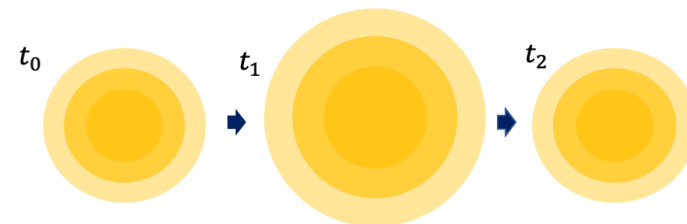
Possible
consequences



Black Hole



Highly perturbed star



Equal mass: ✓

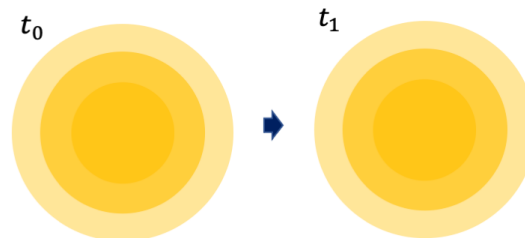
metric correction

$$\gamma_{ij} = \gamma_{ij}^A + \gamma_{ij}^B - \gamma_{ij}^B(x_A^i)$$

$$\gamma_{ij}^B(x_A^i) = \gamma_{ij}^A(x_B^i)$$



Minor
perturbations



Malaise

- premature collapse
- unphysical radial oscillations
- alters GW signal

Helfer, Sperhake et al. CQG2022

Improved superposition

Evstafyeva, Sperhake et al. CQG2023

Unequal mass BS-BS binary: conformal-factor correction

$$\lambda = \chi^{-1} \quad \tilde{\gamma}_{ij} = \lambda^{-1} \gamma_{ij}$$

$$\lambda_{\text{new}}(x^i) = \lambda(x^i) + w_A(x^i)h_A + w_B(x^i)h_B$$

$$\lambda_{\text{new}}(x_A^i) = \lambda(x_A^i) + \delta\lambda(x_A^i) = \lambda_A(x_A^i),$$

$$\lambda_{\text{new}}(x_B^i) = \lambda(x_B^i) + \delta\lambda(x_B^i) = \lambda_B(x_B^i).$$



Correction radius

$$w_J(x^i) = \frac{1}{\sqrt{R_J^2 + r_J^2}}, \quad r_J \equiv \|x^i - x_J^i\|, \quad J \in \{A, B\}$$

$$h_A = \frac{-w_B(x_A^i)\delta\lambda(x_B^i) + w_B(x_B^i)\delta\lambda(x_A^i)}{w_A(x_A^i)w_B(x_B^i) - w_A(x_B^i)w_B(x_A^i)},$$

$$h_B = \frac{w_A(x_A^i)\delta\lambda(x_B^i) - w_A(x_B^i)\delta\lambda(x_A^i)}{w_A(x_A^i)w_B(x_B^i) - w_A(x_B^i)w_B(x_A^i)}.$$

Fos BS-BH binary, problem is more severe and needs careful treatment

$$\delta\lambda_{\text{BH}} = \lambda_{\text{BH}}(x_{\text{BH}}^i) - \lambda(x_{\text{BH}}^i) \rightarrow 0$$

$$R_{\text{BH}} \rightarrow 0$$

→

$$\lambda_{\text{new}}(x^i) = \lambda(x^i) + w_{\text{BS}}(x^i) \frac{\delta\lambda_{\text{BS}}(x_{\text{BS}}^i)}{w_{\text{BS}}(x_{\text{BS}}^i)}$$

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Numerical results

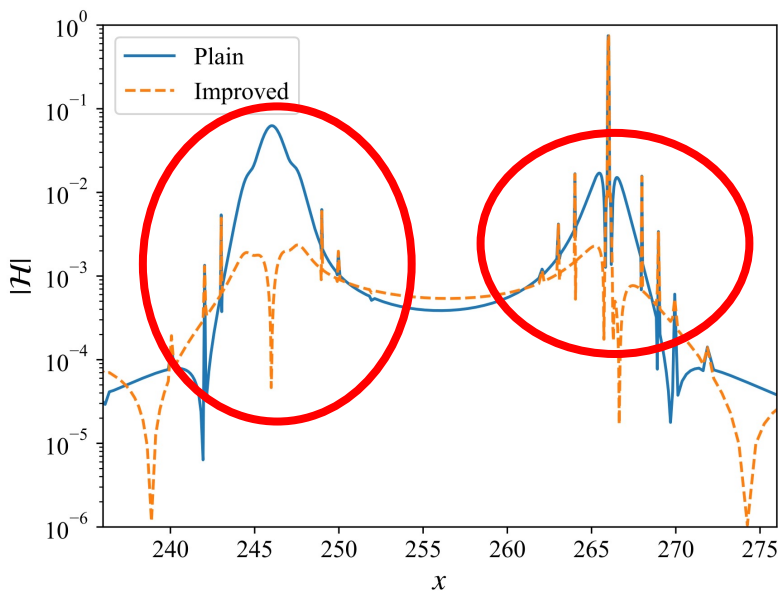
$$V = \mu^2 |\varphi|^2 \left(1 - 2 \frac{|\varphi|^2}{\sigma_0^2}\right)^2 \quad \sigma_0 = 0.2$$

Model	$A(0)$	μM_{BS}	ω/μ	μr_{99}	$\mathcal{C}$	
BS-170	0.17	0.7134	0.4392	3.3224	0.2146	H
BS-147	0.147	0.3609	0.6780	4.2557	0.0848	M
BS-100	0.1	0.2700	0.8506	6.1717	0.0437	L

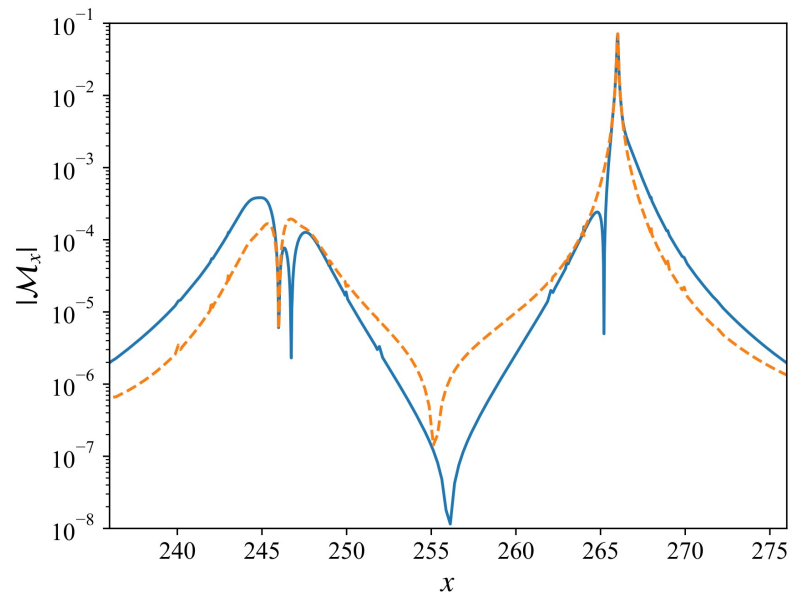
Run	q	Object A	Object B	v_x^A	v_x^B	d	M	Initial data
BSBH-q1-H	1	BS-170	$M_{\text{BH}} = 0.7134$	0.1	-0.1	20, 40, 60, 80	1.4340	Improved/Plain
BSBH-q1-M	1	BS-147	$M_{\text{BH}} = 0.3609$	0.1	-0.1	40	0.7254	Improved
BSBH-q1-L	1	BS-100	$M_{\text{BH}} = 0.2700$	0.1	-0.1	40	0.5427	Improved
BSBH-q038	0.38	BS-100	$M_{\text{BH}} = 0.7134$	0.1447	-0.0553	40	0.9874	Improved
BSBH-q05	0.5	BS-147	$M_{\text{BH}} = 0.7134$	0.1325	-0.0675	40	1.0791	Improved
BSBH-q2	2	BS-170	$M_{\text{BH}} = 0.3609$	0.0675	-0.1325	40	1.0791	Improved
BSBH-q2.64	2.64	BS-170	$M_{\text{BH}} = 0.2700$	0.0553	-0.1447	40	0.9874	Improved
BSBS-q1-H	1	BS-170	BS-170	0.1	-0.1	40	1.4340	Improved
BSBS-q1-M	1	BS-147	BS-147	0.1	-0.1	40	0.7254	Improved
BSBS-q1-L	1	BS-100	BS-100	0.1	-0.1	40	0.5427	Improved
BSBS-q038	0.38	BS-100	BS-170	0.1447	-0.0553	40	0.9874	Improved
BSBS-q05	0.5	BS-147	BS-170	0.1325	-0.0675	40	1.0791	Improved
BHBH-q1	1	$M_{\text{BH}} = 0.7134$	$M_{\text{BH}} = 0.7134$	0.1	-0.1	40	1.4340	Plain
BHBH-q038	0.38	$M_{\text{BH}} = 0.2700$	$M_{\text{BH}} = 0.7134$	0.1447	-0.0553	40	0.9874	Plain
BHBH-q05	0.5	$M_{\text{BH}} = 0.3609$	$M_{\text{BH}} = 0.7134$	0.1325	-0.0675	40	1.0791	Plain

Comparison of initial data

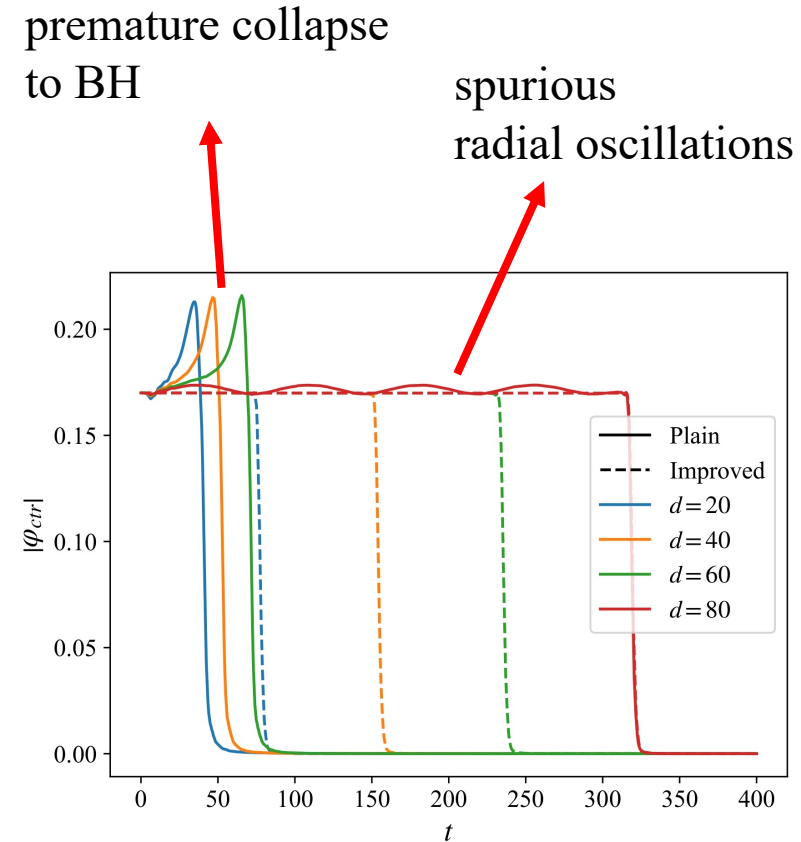
- Initial constraint violations
- Evolution of the scalar field and premature collapse



Initial Hamiltonian constraint

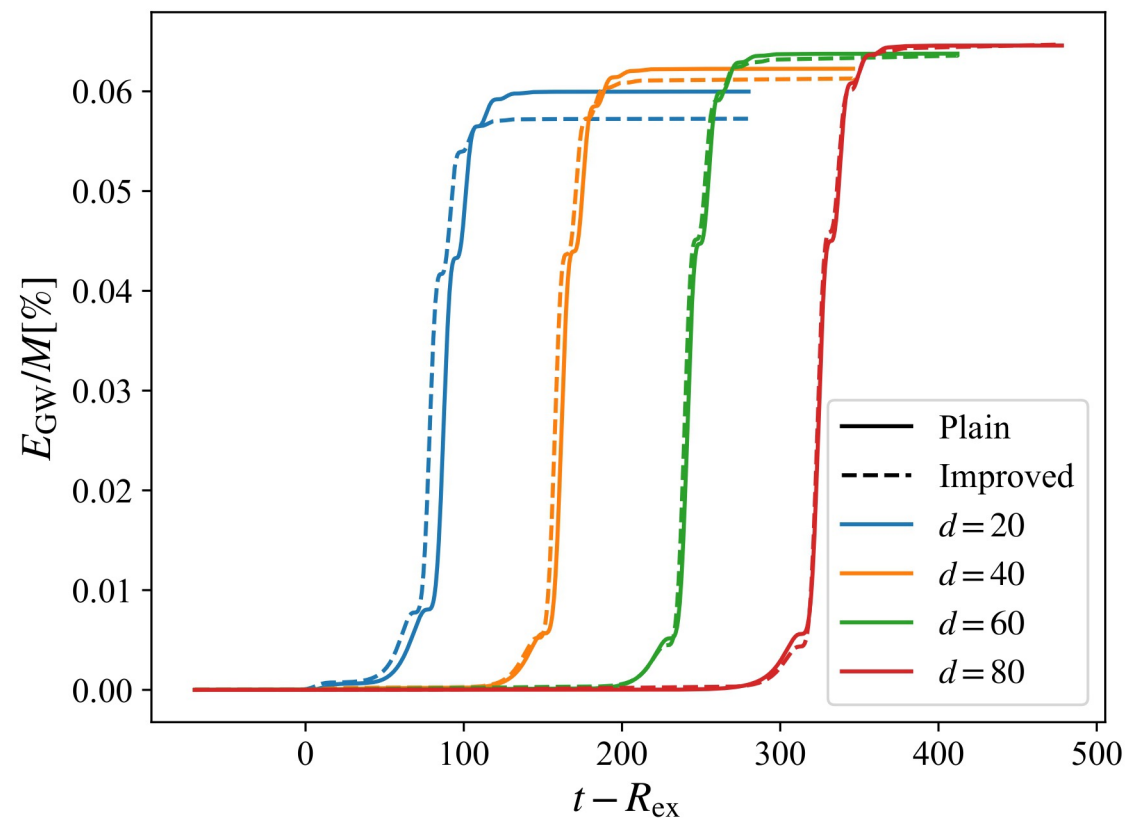


Initial momentum constraint

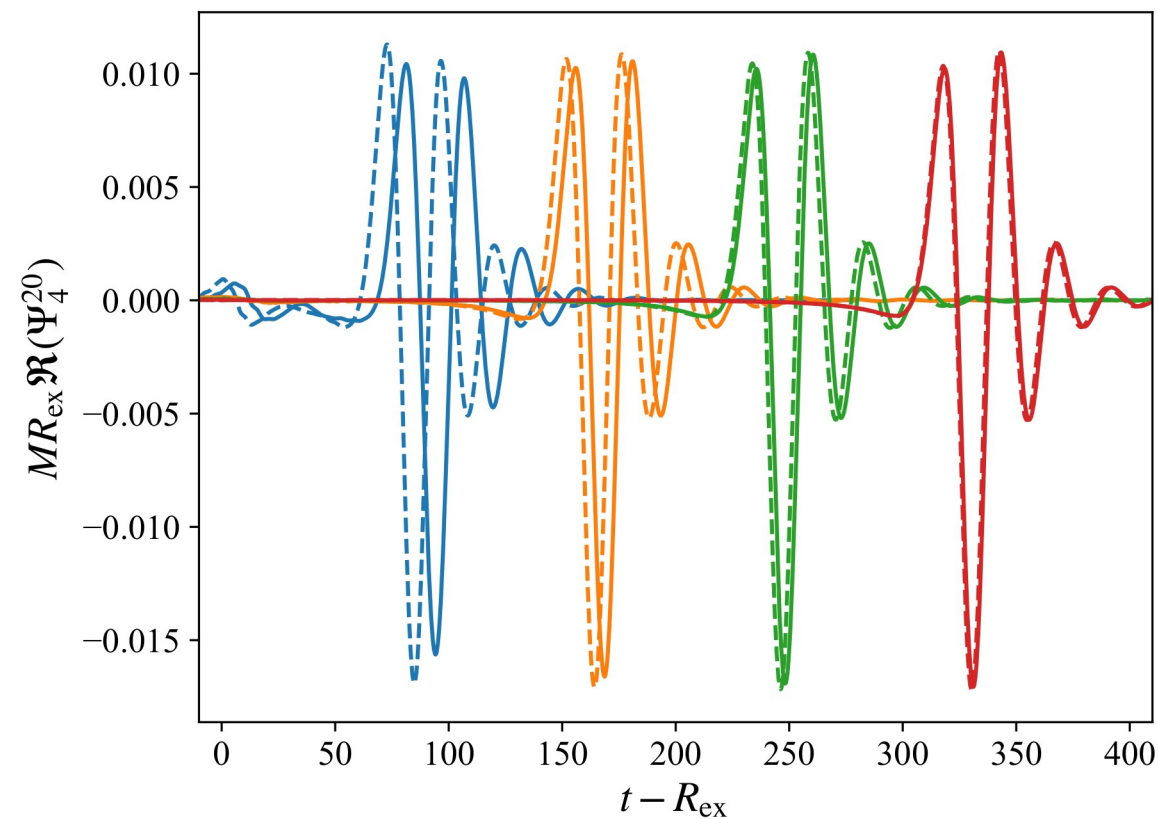


Central scalar-field amplitude

Comparison of initial data

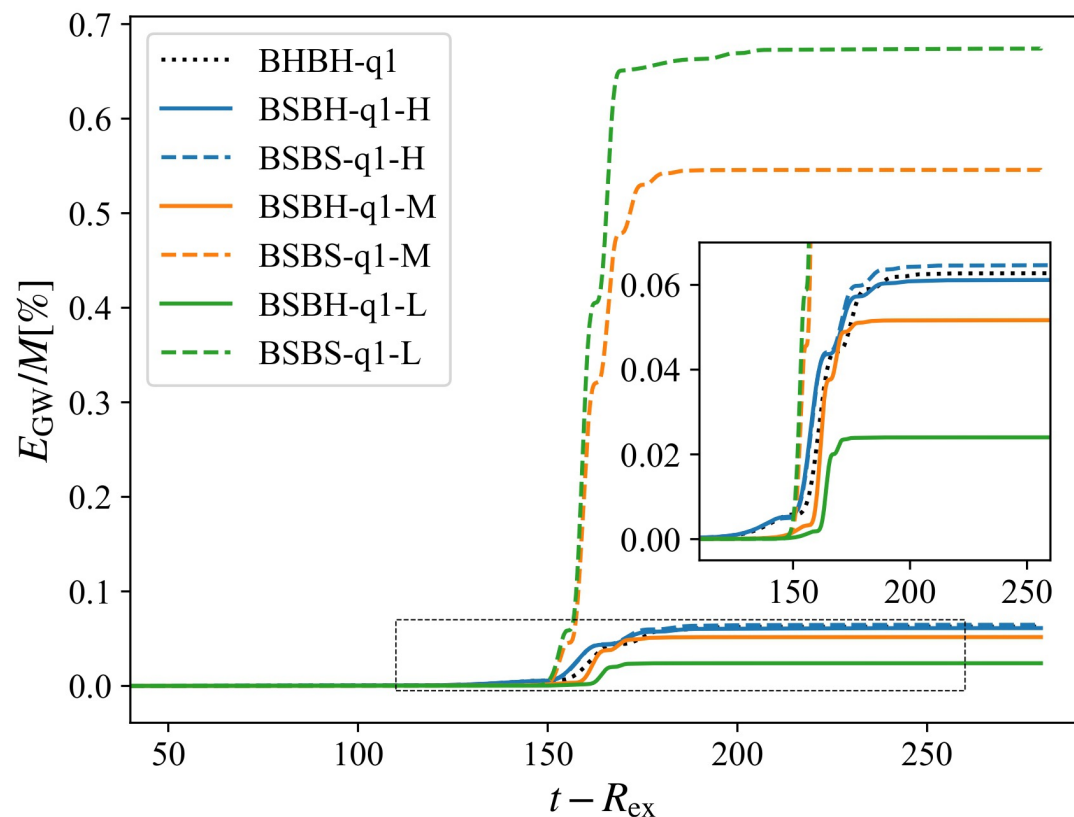


Overestimate GW energy

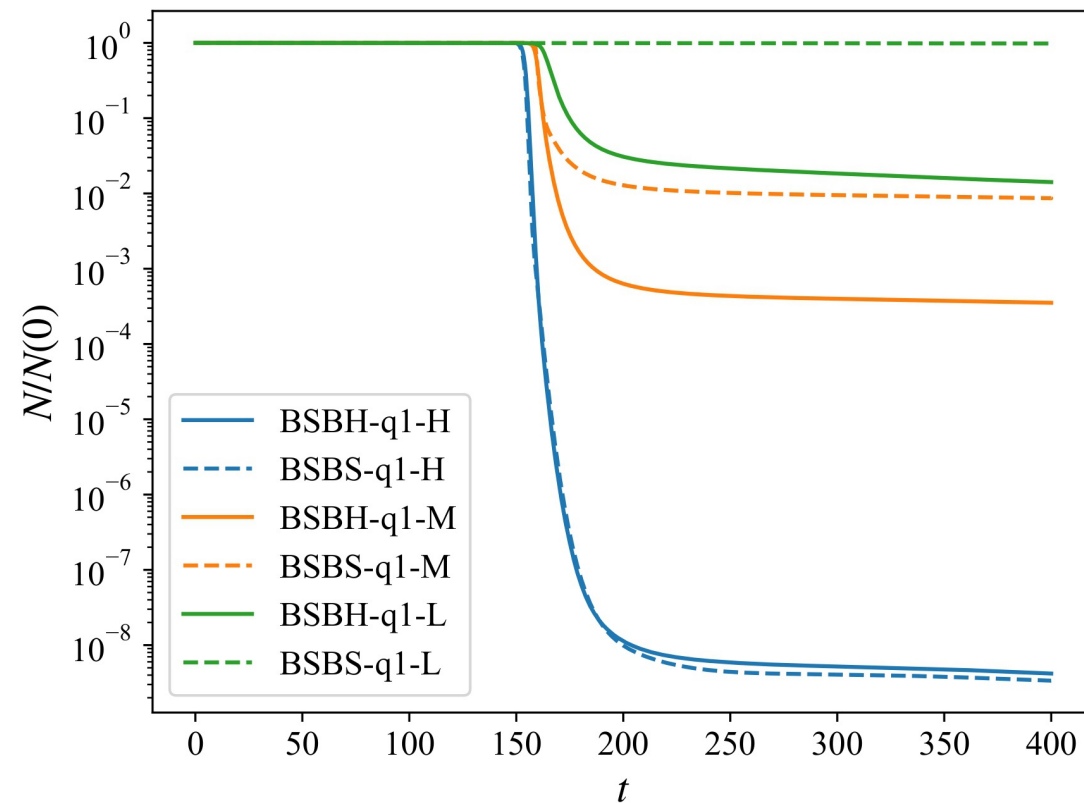


Stretch GW waveform

Equal-mass BS-BH collisions

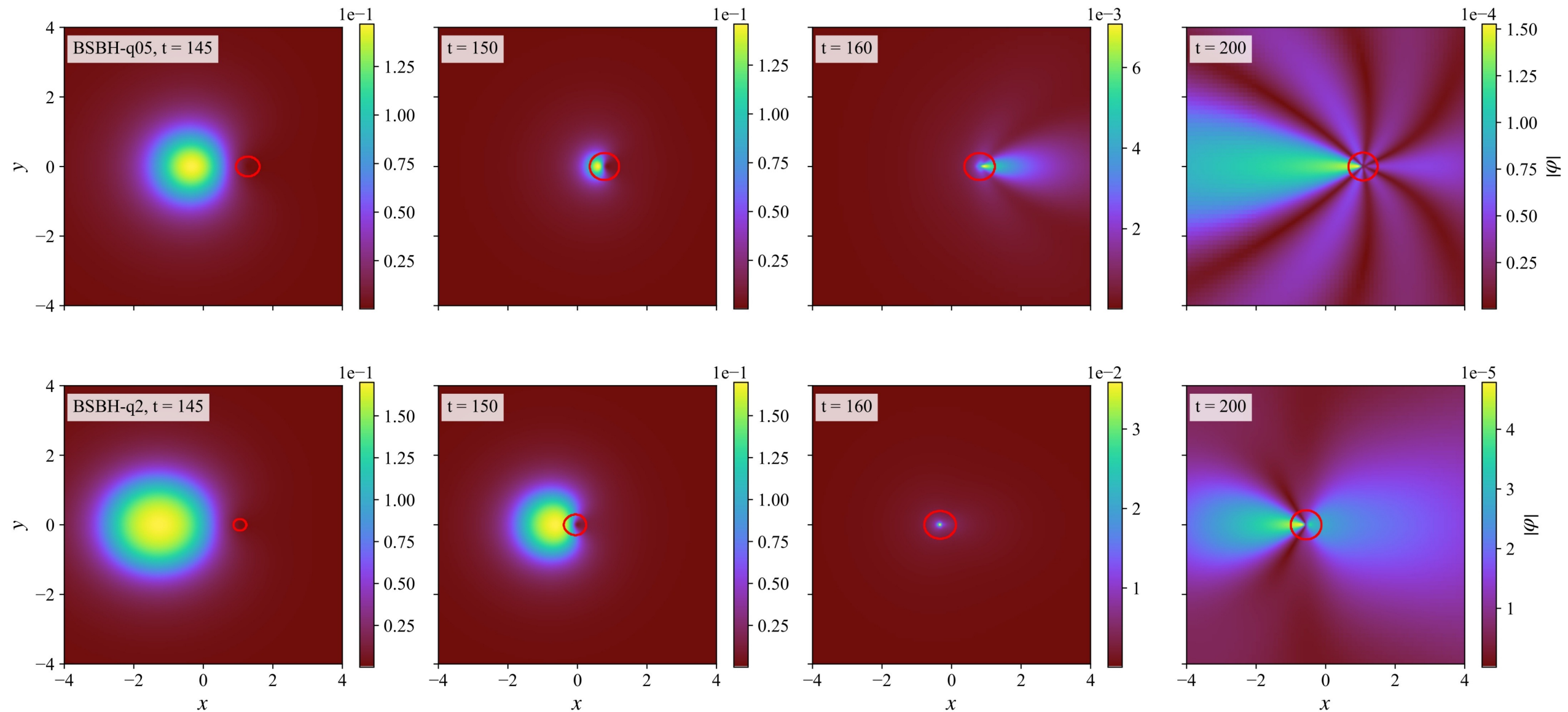


GW radiation efficiency increases
with the compactness of the BS



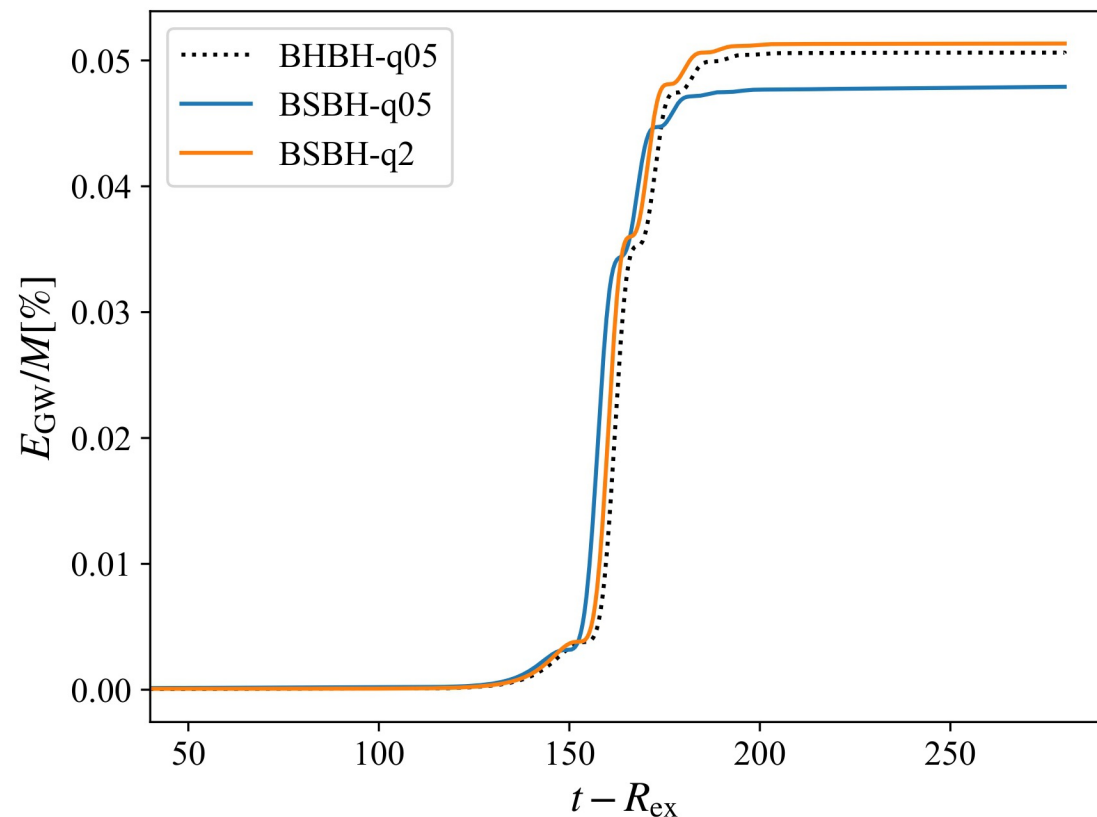
Residual Noether charge $< 10\%$

Unequal-mass BS-BH collisions: $q=0.5$

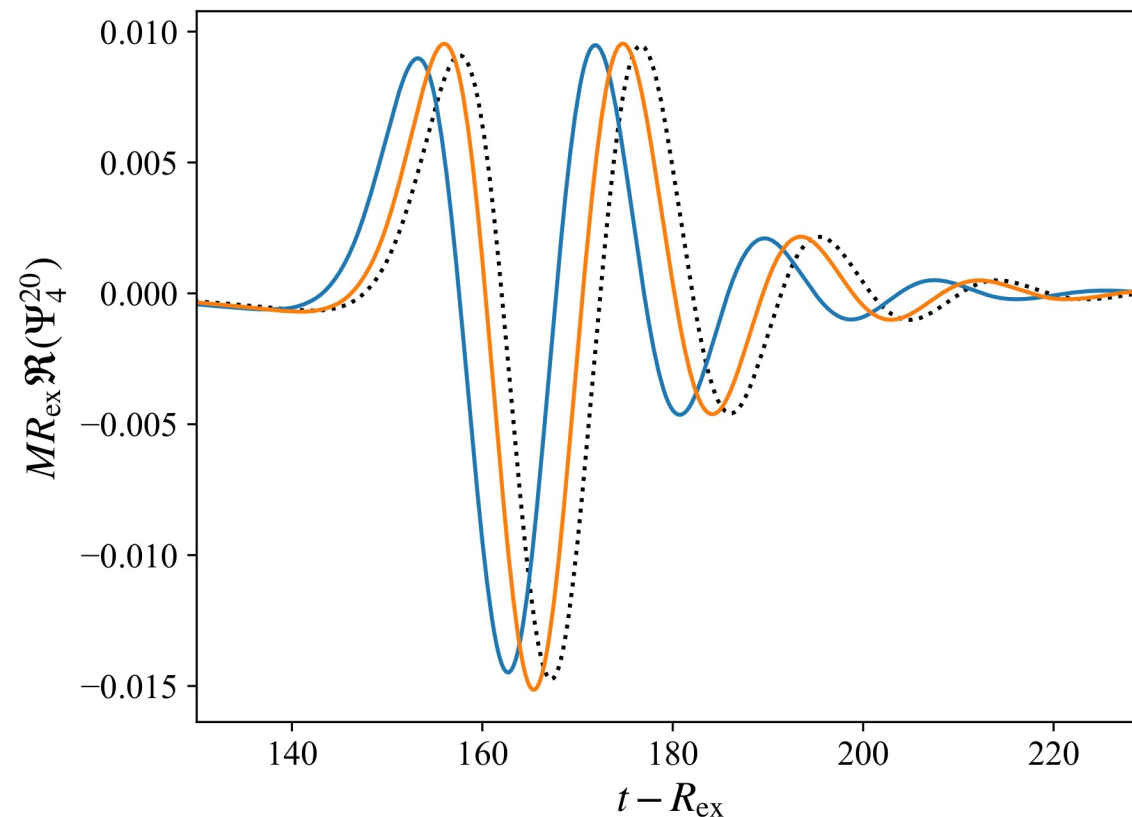


Unequal-mass BS-BH collisions: $q=0.5$

➤ $q = M_{BS}/M_{BH} = 0.5$ or 2



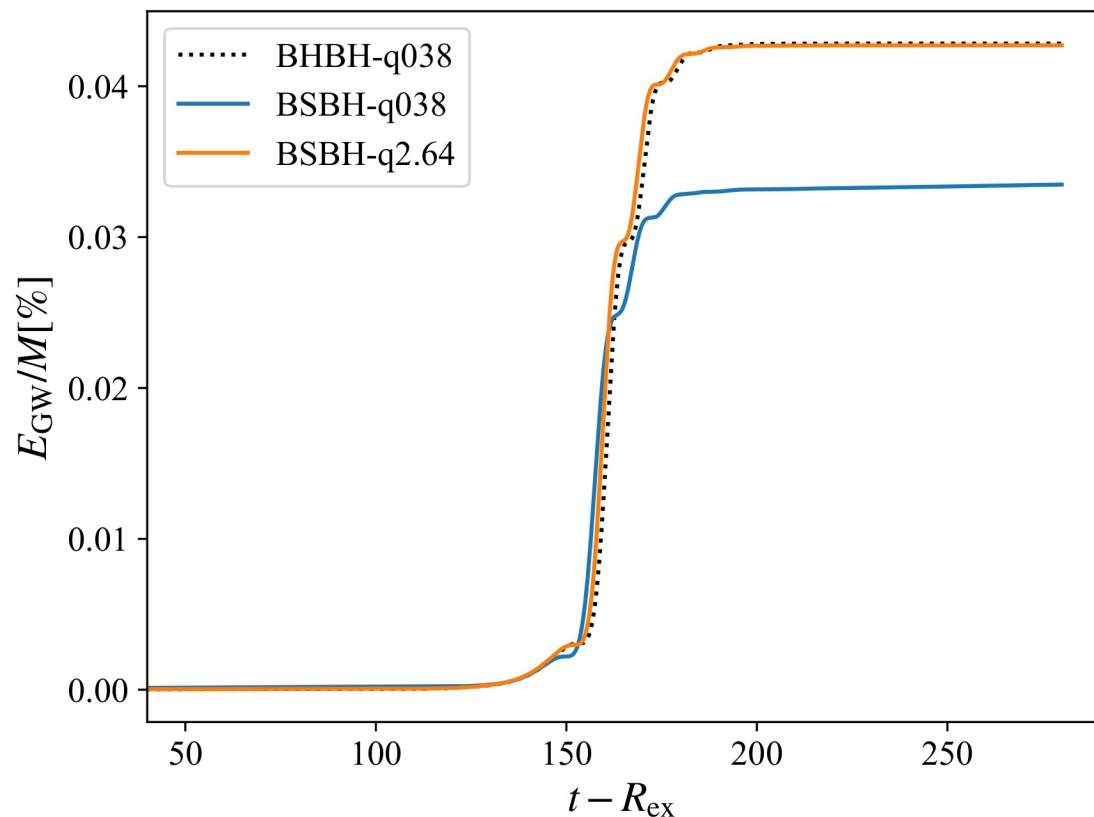
$q = 2$ exceeds BH-BH counterpart



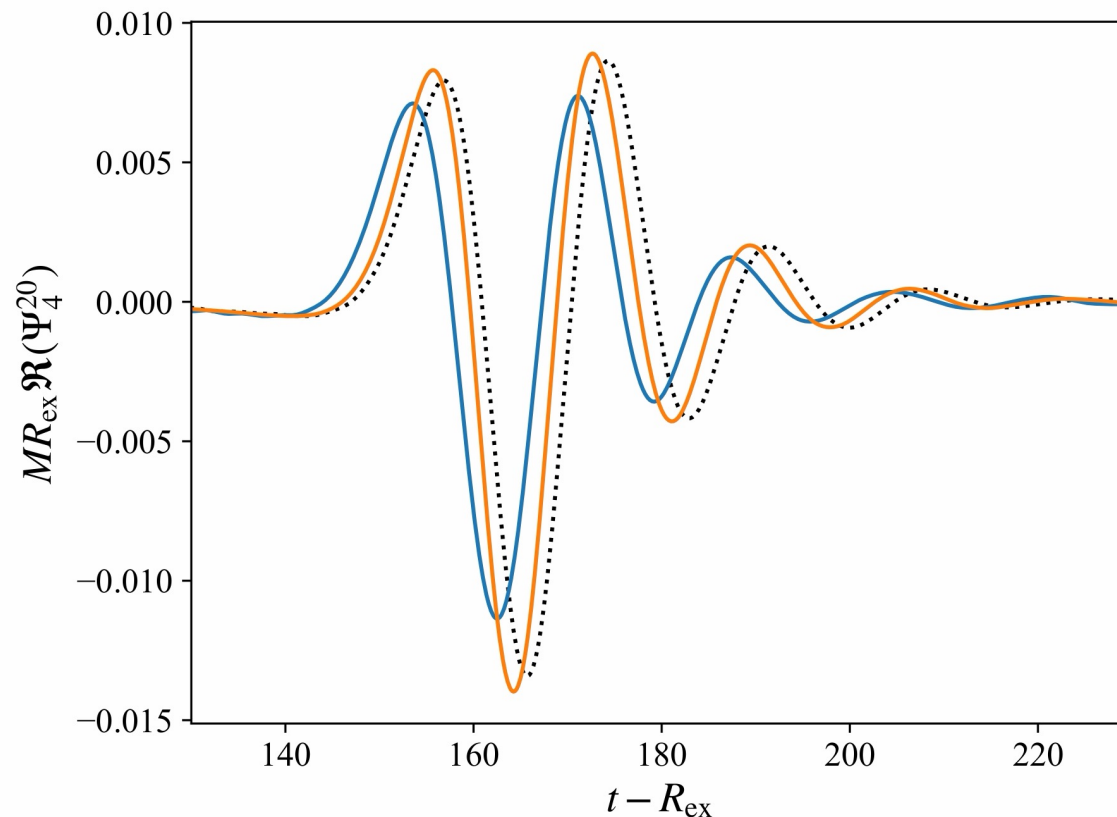
Strong degeneracies in $(2, 0)$ mode

Unequal-mass BS-BH collisions: $q=0.38$

➤ $q = M_{BS}/M_{BH} = 0.38$ or 2.64



$q = 2.64$ is comparable to BH-BH counterpart



Strong degeneracies in $(2, 0)$ mode

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Conclusions and discussions

- Conformal-factor correction improves initial-data quality
 - Suppresses the Hamiltonian constraint violation
 - Removes spurious radial oscillations
 - Avoids premature collapse
- Equal-mass:
 - GW efficiency increases with BS compactness and **approaches BH-BH limit**
 - Remaining scalar field is negligible
- Unequal-mass:
 - $q > 1$ GW efficiency **can exceed BH-BH counterpart**
 - strong degeneracies in (2, 0) mode

Thank you!